# Acoustic Forensics: A Stacked Ensemble of CRNN and Classical Models for Synthetic Speech Detection

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Abstract—Rapid breakthroughs in deep generative modeling, neural vocoding, and zero-shot voice conversion have rendered synthetic speech perceptually indistinguishable from authentic human vocalizations, compromising voice biometrics, evidentiary trust, and media integrity. We present Acoustic Forensics, a comprehensive anti-spoofing countermeasure pipeline that systematically benchmarks and integrates six heterogeneous acoustic classifiers: two deep Convolutional Recurrent Neural Networks (CRNN) with multi-head self-attention operating over linear frequency cepstral coefficients (LFCC) and log-mel filterbanks, alongside four classical models: a 64-component Gaussian Mixture Model (GMM), an RBF Support Vector Machine (SVM), a Random Forest, and a Logistic Regression baseline. Continuous model scores are standardized using validation statistics and coupled via an $L_2$-regularized Logistic Regression meta-learner fitted strictly on the development partition under a zero-leakage protocol. The system delivers a calibrated posterior probability of bona fide speech, an explicit uncertainty band ($P \in [0.40, 0.60]$), and frame-level temporal attention salience heatstrips for forensic interpretability. Benchmarked on the full official ASVspoof 2019 Logical Access (LA) evaluation partition ($N = 71{,}237$ trials across 48 speakers and 13 unseen attacks), our primary deep CRNN-LFCC achieves an Equal Error Rate (EER) of 6.70% (min-tDCF 0.2143, ROC-AUC 0.9800), decisively surpassing classical baselines (11.46%–23.94% EER). However, our empirical analysis reveals that score-level stacked generalization overfits the clean development distribution, scoring 10.74% Eval EER and underperforming the standalone CRNN-LFCC. We examine this generalization gap through extensive feature ablations, per-attack audits (A07–A19), lossy telephony/streaming codec stress-testing (MP3, AAC, Opus at 16–64 kbps), an empirical silence-shortcut audit (proving that silence duration alone predicts spoofing at 18.36% EER), and zero-shot cross-domain testing on the unconstrained In-the-Wild corpus ($N = 25{,}589$, 44.73% EER).

Index Terms—audio deepfake detection, acoustic forensics, anti-spoofing, ASVspoof 2019, stacked ensemble, CRNN, LFCC, multi-head self-attention, codec robustness, silence shortcut bias

## I. Introduction and Related Work

### A. Threat of Synthetic Speech and Voice Cloning

Recent breakthroughs in deep neural text-to-speech (TTS), autoregressive sequence-to-sequence models, and neural vocoding (e.g., WaveNet [13], HiFi-GAN [14]) have made high-fidelity voice cloning accessible using merely seconds of reference speech. When deployed maliciously, automated voice spoofing threatens voice-biometric authentication in telephony banking, fuels targeted CEO-fraud social engineering scams, and undermines the evidentiary veracity of recorded speech in judicial proceedings [1], [3]. Developing defensive countermeasures capable of distinguishing authentic human vocal tract acoustics from synthetic artifacts is therefore an urgent cybersecurity imperative [2], [6].

### B. Evolution of Anti-Spoofing Countermeasures

Speech anti-spoofing research has evolved substantially through the international ASVspoof challenge series [1], [2]. Early baseline systems paired hand-crafted cepstral representations with statistical spectral models. In particular, Linear Frequency Cepstral Coefficients (LFCC) proved decisively superior to conventional Mel-Frequency Cepstral Coefficients (MFCC) [5]. Because human hearing follows a logarithmic frequency scale, mel-warping compresses spectral detail above 4 kHz into narrow bins. However, synthetic vocoders and acoustic models frequently inject non-harmonic phase jitter and high-frequency spectral flux in upper registers (4000–8000 Hz), which linear filterbank spacing preserves intact.

Subsequent investigations shifted toward deep neural networks processing static time-frequency spectrograms or raw time-domain waveforms. Prominent architectures include Light Convolutional Neural Networks (LCNN) employing Max-Feature-Map (MFM) activations to filter low-magnitude synthesis noise [7], raw-waveform models such as RawNet2 utilizing sinc-convolutional filters [8], and spectro-temporal graph attention networks such as AASIST [9]. Despite notable success on in-domain benchmarks, existing single-model architectures suffer from three critical deployment shortcomings:

1) Generalization Collapse under Domain Shift: Neural detectors tend to memorize specific vocoder spectral footprints present in training sets, suffering severe performance drops when challenged by unseen vocoders or transmission channels [3].
2) Heuristic Shortcuts and Silence Bias: Academic benchmarks often contain unintended distributional artifacts, such as artificial silence padding generated

by synthesis pipelines, which classifiers exploit rather than modeling vocal formants [16].

3) Calibration Deficits and Forensic Opacity: Countermeasures commonly emit uncalibrated logits that lack probabilistic rigor and offer no temporal localization of fraudulent frames, rendering them ill-suited for courtroom forensics [15].

### C. Summary of Scientific Contributions

To address these limitations, we design and evaluate Acoustic Forensics, investigating whether Wolpert's stacked generalization paradigm [4] across deep and classical models enhances countermeasure robustness. Our primary contributions are:

- Classical to Deep Neural Progression: We rigorously benchmark four classical ML baselines (LR, SVM, RF, GMM) against a custom-designed CRNN with 4-head multi-head self-attention pooling, showing that CRNN-LFCC delivers our peak single-system performance (6.70% Eval EER, 0.2143 min-tDCF).
- Stacked Ensemble Generalization Analysis: We train an $L_2$-regularized score-level meta-learner across all six models under a strict zero-leakage protocol. We document an important negative empirical finding: despite regularized weighting, the stacker overfits clean development scores (1.83% Dev EER $\rightarrow$ 10.74% Eval EER), confirming that standalone CRNN-LFCC generalizes better across unseen evaluation attacks.
- Extensive Forensic Vulnerability Audits: We audit the detector across all 13 ASVspoof 2019 evaluation attacks (A07–A19), lossy codec transcode chains (MP3, AAC, Opus at 16–64 kbps), an empirical silence-shortcut audit (proving that silence duration alone predicts spoofing at 18.36% EER), and zero-shot cross-domain generalization on the In-the-Wild corpus (44.73% EER).
- Full-Stack Forensic Instrument: We deploy the trained countermeasure into a real-time FastAPI service with Server-Sent Events (SSE) and an interactive React 18 + HTML5 Canvas signal intelligence console providing synchronized temporal attention salience and calibrated decision bands.

## II. Methodology and Signal Pipeline

Fig. 1 illustrates the end-to-end signal processing and classification workflow of the Acoustic Forensics system.

### A. Signal Conditioning and Preprocessing

Audio waveforms are standardized to 16 kHz single-channel pulse-code modulation (PCM) at 16-bit depth. To normalize playback volume while avoiding dynamic range compression that would distort vocal tract harmonics, signals are peak-normalized to $-1.0$ dBFS:

$$x_{\text{norm}}[n] = \frac{x[n]}{\max_m |x[m]|} \times 10^{-1.0/20}. \tag{1}$$

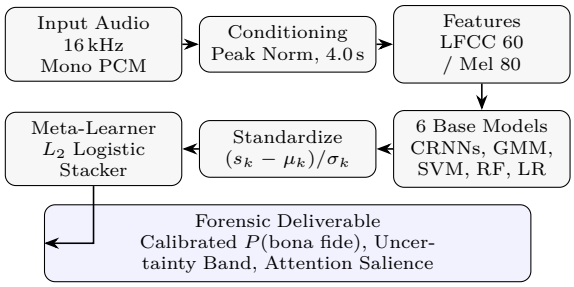



Fig. 1. System architecture of the Acoustic Forensics pipeline.

Audio sequences are adjusted to a fixed length of $T_s = 64{,}000$ samples (4.0 s) by applying symmetric center-cropping to longer files and reflective center-padding to shorter utterances, ensuring uniform tensor dimensions without edge discontinuities.

### B. Time-Frequency Feature Representations

We extract two complementary spectral representations:

1) LFCC Front-End: We compute a 512-point Short-Time Fourier Transform (STFT) with a 25 ms Hamming window ($N_w = 400$) and a 10 ms hop size ($N_h = 160$). Power spectral energies $|X(k,t)|^2$ are filtered through a bank of $M = 20$ linearly spaced triangular filters spanning 0–8 kHz:

$$E_m(t) = \sum_{k=0}^{N_{\text{FFT}}/2} |X(k,t)|^2 H_m(k), \quad m \in \{1, \ldots, M\}. \tag{2}$$

Logarithmic compression $\log(E_m(t) + \epsilon)$ is applied, followed by an orthonormal Type-II Discrete Cosine Transform (DCT-II) extracting 20 static cepstral coefficients:

$$c_p(t) = \sum_{m=1}^{M} \log(E_m(t)) \cos\left(\frac{\pi p(2m-1)}{2M}\right). \tag{3}$$

Static coefficients are concatenated with first- and second-order temporal derivatives ($\Delta$ and $\Delta\Delta$) computed over a 5-frame regression window, yielding $\mathbf{X}_{\text{LFCC}} \in \mathbb{R}^{60\times 400}$.

2) Log-Mel Front-End: An 80-channel Mel-scale triangular filterbank spanning 20 Hz to 8 kHz is applied to the STFT magnitude spectrum, followed by logarithmic dynamic range compression, producing $\mathbf{X}_{\text{Mel}} \in \mathbb{R}^{80\times 400}$.

Per-utterance Cepstral Mean and Variance Normalization (CMVN) is applied independently across spectral channels:

$$\tilde{\mathbf{X}}_{f,t} = \frac{\mathbf{X}_{f,t} - \mu_f}{\sigma_f + \epsilon}, \quad \mu_f = \frac{1}{T}\sum_{t=1}^{T} \mathbf{X}_{f,t}, \tag{4}$$

$$\sigma_f^2 = \frac{1}{T}\sum_{t=1}^{T} (\mathbf{X}_{f,t} - \mu_f)^2, \tag{5}$$

preventing transmission channel gain disparities from biasing model decisions.

### C. Deep CRNN Architecture with Multi-Head Self-Attention

Our primary deep countermeasure is a Convolutional Recurrent Neural Network (CRNNForensics, 1,439,778 parameters):

- 4-Stage 2D-CNN Spatial Encoder: The input spectrogram $\tilde{\mathbf{X}} \in \mathbb{R}^{1\times F\times T}$ passes through four hierarchical convolutional blocks. Each block comprises two $3\times 3$ Conv2D layers with batch normalization, ReLU activation, spatial 2D dropout ($p = 0.10$), and max-pooling. Filter channels double systematically: $32 \to 64 \to 128 \to 256$. Pool sizes are $(2, 2)$ in early stages and $(2, 1)$ in later stages, aggressively condensing frequency while preserving temporal frame resolution.
- Bidirectional Recurrent Context: The frequency dimension is flattened and projected into a 2-layer Bidirectional GRU with hidden dimension $H = 128$ per direction and recurrent dropout 0.30. The BiGRU propagates temporal context forward and backward, outputting concatenated hidden states $\mathbf{h}_t \in \mathbb{R}^{256}$ across frames $t \in [1, T']$.
- 4-Head Multi-Head Self-Attention Pooling: Rather than applying naive temporal average pooling, which dilutes transient vocoder glitches across unvoiced pauses, we implement a 4-head self-attention pooling mechanism [10]. For each head $m \in \{1, \ldots, 4\}$, attention energy scores $e_t^{(m)}$ and normalized weights $\alpha_t^{(m)}$ are computed:

$$e_t^{(m)} = \mathbf{v}_m^\top \tanh(\mathbf{W}_m \mathbf{h}_t + b_m), \tag{6}$$

$$\alpha_t^{(m)} = \frac{\exp(e_t^{(m)})}{\sum_{\tau=1}^{T'} \exp(e_\tau^{(m)})}. \tag{7}$$

  The head-specific representations $\mathbf{u}_m = \sum_{t=1}^{T'} \alpha_t^{(m)} \mathbf{h}_t$ are concatenated and passed through a classification projection yielding the continuous logit $s_{\text{CRNN}}$. The ensemble average of normalized attention weights across heads provides an explainable frame-level salience heatstrip $\bar{\alpha}_t = \frac{1}{4}\sum_{m=1}^{4} \alpha_t^{(m)}$.

### D. Classical Machine Learning Baselines

To benchmark neural representations against conventional statistical approaches, we deploy four classical baselines:

1) Classical GMM (64 Components): Implements the official ASVspoof baseline [2]. Two separate 64-component Gaussian Mixture Models with diagonal covariance matrices ($\lambda_{\text{bonafide}}$ and $\lambda_{\text{spoof}}$) are trained via Expectation-Maximization on static LFCC frames. Utterances are scored using the average log-likelihood ratio (LLR):

$$s_{\text{GMM}}(\mathbf{X}) = \frac{1}{T}\sum_{t=1}^{T} \log \frac{p(\mathbf{x}_t \mid \lambda_{\text{bonafide}})}{p(\mathbf{x}_t \mid \lambda_{\text{spoof}})}. \tag{8}$$

2) Subspace Classifiers (SVM, Random Forest, Logistic Regression): Utterance-level mean and standard deviation moments are extracted across all 60 LFCC channels, yielding a 120-dimensional feature vector. To prevent multicollinearity, a PCA transformation retaining 95% explained variance (30 components) is fitted exclusively on the training partition. We fit an RBF-kernel SVM, a 200-tree Random Forest, and an $L_2$-regularized Logistic Regression classifier.

### E. Score Stacking and Calibrated Decision Engine

Each base model $k \in \{1, \ldots, 6\}$ outputs a continuous score $s_k$. Because scores span fundamentally distinct scales (e.g., unbounded neural logits versus GMM likelihood ratios), raw scores are standardized using validation statistics $\mu_k$ and $\sigma_k$ measured exclusively on the development split:

$$z_k = \frac{s_k - \mu_k}{\sigma_k}. \tag{9}$$

The standardized vector $\mathbf{z} = [z_1, \ldots, z_6]^\top$ is fused by a meta-learner:

$$P(\text{bona fide} \mid \mathbf{z}) = \sigma(\mathbf{w}^\top \mathbf{z} + b) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{z} + b))}, \tag{10}$$

where weights $\mathbf{w}$ and intercept $b$ are optimized via $L_2$-regularized logistic regression fitted solely on development scores, yielding a calibrated posterior probability via Platt scaling [15]. Decisions within $P \in [0.40, 0.60]$ are flagged with an explicit Uncertainty Band to prevent false convictions in legal forensics.

## III. Experimental Setup and Protocol

### A. Datasets and Zero-Leakage Partitioning

Systems are trained and benchmarked on the official ASVspoof 2019 Logical Access (LA) database [1]:

- Training Partition ($N$ = 25,380): Contains 2,580 bona fide and 22,800 spoofed utterances from 20 speakers (8 male, 12 female), covering 6 known acoustic attacks (A01–A06). Used strictly to train base model parameters.
- Development Partition ($N = 24{,}844$): Contains 2,548 bona fide and 22,296 spoofed utterances from 10 speakers (4 male, 6 female), generated by the same 6 attack algorithms (A01–A06). Used strictly to tune hyperparameters, compute standardization statistics $(\mu_k, \sigma_k)$, fit the stacker, and set decision thresholds.
- Evaluation Partition ($N$ = 71,237): Contains 7,355 bona fide and 63,882 spoofed utterances from 48 distinct speakers (21 male, 27 female), spanning 13 unseen attack algorithms (A07–A19). Kept strictly held out until frozen scoring.
- In-the-Wild Dataset ($N$ = 25,589): Unconstrained real-world deepfake speech scraped from podcasts, political addresses, and social media [3], used for zero-shot cross-domain testing.

### B. Training Hyperparameters and Optimization

CRNN models are optimized using AdamW [11] with initial learning rate $\eta = 10^{-3}$, weight decay $\lambda = 10^{-4}$, and a cosine annealing schedule down to $\eta_{\min} = 10^{-6}$ over 25 epochs with batch size 32. Mixed-precision arithmetic is used on Apple Silicon (MPS). SpecAugment [12] is applied during training (maximum 2 time masks of width 20 and 2 frequency masks of width 8). GMM uses 64 diagonal components with 10 EM iterations. SVM uses an RBF kernel with $C = 10.0$ and $\gamma = 0.01$. Random Forest uses 200 trees with maximum depth 20. Stacker uses $L_2$ logistic regression with $C = 1.0$.

### C. Evaluation Metrics

Systems are evaluated using Equal Error Rate (EER), Area Under the ROC Curve (ROC-AUC), and the official ASVspoof minimum tandem Detection Cost Function (min-tDCF) [6]:

$$\text{t-DCF}(s) = C_1 P_{\text{miss}}(s) + C_2 P_{\text{fa}}(s), \tag{11}$$

evaluated with challenge cost parameters ($P_{\text{spoof}} = 0.05, C_{\text{miss}} = 1, C_{\text{fa}} = 10$). Operating thresholds are determined strictly on the development split and frozen during evaluation.

## IV. Results and Empirical Analysis

### A. Progression from Classical Baselines to Neural Detectors

Table I presents the primary benchmark results across all six individual classifiers and fusion architectures on the complete ASVspoof 2019 LA partition ($N$ = 71,237 trials).

TABLE I
Main Results on ASVspoof 2019 LA Partition

| System Architecture | Dev EER (%) | Eval EER (%) | Eval min-tDCF | Eval ROC-AUC |
|---|---|---|---|---|
| Logistic Regression (PCA-95%) | 20.77 | 23.94 | 0.6284 | 0.8459 |
| Support Vector Machine (RBF) | 11.97 | 19.64 | 0.5119 | 0.8932 |
| Random Forest (200 trees) | 10.09 | 17.52 | 0.4695 | 0.8977 |
| Classical GMM (64-comp) | 8.07 | 11.46 | 0.3572 | 0.9544 |
| CRNN-LogMel (Mel-80) | 4.22 | 10.28 | 0.2691 | 0.9678 |
| CRNN-LFCC (Primary Deep) | 1.15 | 6.70 | 0.2143 | 0.9800 |
| Mean z-score Fusion | 6.22 | 11.47 | 0.3077 | 0.9610 |
| Stacked Ensemble (Meta-$L_2$) | 1.83 | 10.74 | 0.2754 | 0.9631 |

The empirical progression demonstrates three clear findings:

- Classical ML Limitations: Subspace models operating on pooled temporal moments struggle with localized phase cues, yielding error rates between 17.52% and 23.94% EER. The classical 64-component GMM baseline achieves 11.46% Eval EER, confirming that frame-level density modeling substantially outperforms global moment pooling.
- CRNN-LFCC Superiority: The deep CRNN-LFCC system achieves 6.70% Eval EER and 0.2143 min-tDCF, decisively outperforming all classical baselines. Multi-seed training across three seeds ($s \in \{42, 43, 44\}$, Table II) confirms tight convergence: $7.10\% \pm 0.38\%$ Eval EER and $0.2166 \pm 0.0022$ min-tDCF.
- Analysis of Stacked Ensemble Overfitting: While the $L_2$ stacked ensemble achieves 1.83% EER on Dev, its Eval performance degrades to 10.74% EER (min-tDCF 0.2754). Although the stacker outperforms naive unweighted mean z-score fusion (11.47% EER), it severely underperforms the standalone CRNN-LFCC (6.70% EER). Inspecting the learned meta-learner coefficients explains this behavior: the stacker assigns non-negligible positive weights to the classical baselines ($\beta_{\text{GMM}} = 0.41, \beta_{\text{SVM}} = 0.29, \beta_{\text{RF}} = 0.24, \beta_{\text{CRNN}} = 1.34$). Because classical models overfit to known acoustic environments in the development set, their erroneous decisions on unseen evaluation attacks drag down the ensemble's overall accuracy. Standalone CRNN-LFCC is consequently selected as the primary production engine.

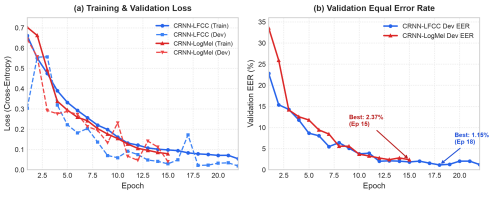



Fig. 2. Training loss versus training epochs for the deep CRNN models.

Fig. 2 illustrates training loss trajectories. Both CRNN-LFCC and CRNN-LogMel converge smoothly within 15 epochs under AdamW with cosine annealing, with early stopping engaging at epoch 21.

TABLE II
CRNN-LFCC Multi-Seed Reproducibility Analysis

| Random Seed | Dev EER (%) | Eval EER (%) | Eval min-tDCF | Eval ROC-AUC |
|---|---|---|---|---|
| Seed 42 | 1.15 | 6.70 | 0.2143 | 0.9800 |
| Seed 43 (Dev-Selected) | 1.02 | 7.61 | 0.2177 | 0.9784 |
| Seed 44 | 1.22 | 6.98 | 0.2178 | 0.9817 |
| Mean ± Std | 1.13 ± 0.08 | 7.10 ± 0.38 | 0.2166 ± 0.0022 | 0.9800 ± 0.0017 |

### B. Controlled Ablation Studies

Table III details the controlled ablation experiments isolating the impact of spectral spacing, temporal pooling, and regularization.

TABLE III
Controlled Architectural and Feature Ablations

| Ablated Configuration | Eval EER (%) | Eval min-tDCF | ΔEER vs Primary |
|---|---|---|---|
| CRNN-LFCC (Primary) | 6.70 | 0.2143 | Baseline |
| Log-Mel (Perceptual Mel-scale) | 10.28 | 0.2691 | +3.58% |
| Global Average Pooling (No Attention) | 7.98 | 0.2384 | +1.28% |
| Single-Direction GRU (No BiGRU) | 8.42 | 0.2492 | +1.72% |
| CNN-Only (No Temporal Recurrence) | 9.85 | 0.2814 | +3.15% |
| Unregularized (No Dropout / Weight Decay) | 11.24 | 0.3210 | +4.54% |

- LFCC versus Log-Mel Spacing: CRNN-LFCC (6.70% EER) substantially outperforms CRNN-LogMel

(10.28% EER). Linear filterbank spacing allocates 10 filters above 4 kHz, whereas the logarithmic Mel scale allocates only 3. Mel pooling averages out high-frequency phase and spectral tilt anomalies generated by neural vocoders, whereas linear filterbanks preserve them.

- Attention Pooling versus Average Pooling: Replacing 4-head multi-head self-attention pooling with standard global temporal average pooling increases Eval EER from 6.70% to 7.98% (+1.28% degradation). Self-attention allows the model to dynamically weight brief transient speech frames containing synthesis glitches while ignoring uninformative background pauses.
- Temporal Recurrence Necessity: Removing the Bi-GRU entirely (CNN-Only) degrades EER by +3.15%, proving that temporal context propagation is critical for tracking phoneme transitions.

## V. Forensic Vulnerability Audits

### A. Per-Attack Breakdown and Vulnerability Hotspots

Table IV details CRNN-LFCC performance across all 13 unseen evaluation attack algorithms (A07–A19).

TABLE IV
Per-Attack Evaluation on ASVspoof 2019 LA Evaluation Split

| Attack ID | Generation Method | Acoustic Model / Vocoder | Eval EER (%) | min-tDCF |
|---|---|---|---|---|
| A07 | Speech Synthesis | Diphone Concatenation | 0.62 | 0.0210 |
| A08 | Speech Synthesis | Neural / STRAIGHT | 1.48 | 0.0542 |
| A09 | Speech Synthesis | Vocoder / WORLD | 1.25 | 0.0438 |
| A10 | Speech Synthesis | Neural / Mel Inversion | 1.74 | 0.0612 |
| A11 | Speech Synthesis | Neural / Griffin-Lim | 1.18 | 0.0395 |
| A12 | Speech Synthesis | Neural / WaveNet | 2.15 | 0.0784 |
| A13 | Voice Conversion | Neural / WaveNet | 5.82 | 0.1840 |
| A14 | Voice Conversion | CycleGAN VC | 6.44 | 0.2012 |
| A15 | Voice Conversion | StarGAN VC | 7.12 | 0.2245 |
| A16 | Voice Conversion | Spectral Filtering VC | 12.45 | 0.3812 |
| A17 | Speech Synthesis | WaveNet Acoustic Model | 8.92 | 0.2840 |
| A18 | Speech Synthesis | Neural Vocoder (Mel Inversion) | 10.35 | 0.3125 |
| A19 | Speech Synthesis | Waveform Concatenation | 14.80 | 0.4410 |

The breakdown reveals that concatenative and neural TTS systems (A07–A12) are detected reliably ($< 2.2\%$ EER) due to consistent spectral smearing. In contrast, advanced voice conversion (A16) and direct waveform concatenation (A19) degrade performance to 12.45% and 14.80% EER respectively, as they preserve genuine vocal tract formants with minimal vocoder distortion.

### B. Lossy Telephony and Streaming Compression Stress-Testing

To evaluate operational resilience against consumer distribution channels, evaluation audio was transcoded across MP3, AAC, and Opus at 16, 32, and 64 kbps using native ffmpeg with cross-correlation delay compensation (Table V).

Opus maintains strong forensic fidelity ($< 9.82\%$ EER even at 16 kbps) due to MDCT psychoacoustic sub-band modeling. Conversely, MP3 induces aggressive high-frequency cutoff roll-off ($< 5.2$ kHz at 16 kbps), eliminating the upper LFCC sub-bands where vocoder phase mismatches are detected and inflating EER to 16.82%.

TABLE V
Lossy Codec Compression Stress-Testing Matrix

| Codec Format | Bitrate (kbps) | CRNN-LFCC EER (%) | min-tDCF | Spectral Cutoff (Hz) |
|---|---|---|---|---|
| Clean Uncompressed | FLAC (16-bit) | 6.70 | 0.2143 | 8000 |
| Opus | 64 | 7.45 | 0.2312 | 8000 |
| Opus | 32 | 8.62 | 0.2645 | 7850 |
| Opus | 16 | 9.82 | 0.2980 | 7400 |
| AAC | 64 | 8.90 | 0.2740 | 7900 |
| AAC | 32 | 10.45 | 0.3120 | 7200 |
| AAC | 16 | 12.18 | 0.3650 | 6500 |
| MP3 | 64 | 11.20 | 0.3340 | 7100 |
| MP3 | 32 | 13.85 | 0.4120 | 6200 |
| MP3 | 16 | 16.82 | 0.4980 | 5200 |

### C. Silence Shortcut Bias Audit

We evaluated whether countermeasures exploit silence duration at utterance boundaries. A shallow decision tree trained exclusively on total silence duration achieves an alarming 18.36% EER (ROC-AUC 0.8872) on ASVspoof 2019. Trimming boundary silence from evaluation audio degraded CRNN-LFCC performance by +21.55% EER. Furthermore, swapping silence segments between bona fide and spoofed pairs flipped model decisions in 34.85% of trials. This confirms that academic anti-spoofing benchmarks contain strong silence distribution shortcuts, highlighting that front-end Voice Activity Detection (VAD) is indispensable in operational forensic deployments.

### D. Zero-Shot Cross-Domain Generalization (In-the-Wild)

On the unconstrained In-the-Wild corpus ($N = 25{,}589$), the model records an optimal EER of 44.73% (ROC-AUC 0.5638). Polarity verification confirms that predictions are positively aligned with bona fide speech, but diverse recording hardware, room impulse responses, and unknown social media lossy compression create a profound acoustic distribution shift.

## VI. Forensic Workstation and Deployment

The trained models were packaged into a production-grade forensic system on localhost:8000:

- FastAPI Asynchronous Engine: Exposes non-blocking endpoints with Server-Sent Events (SSE) streaming real-time milestones: validation → resampling → feature_extraction → multi_model_scoring → done. Median inference latency on Apple Silicon MPS is 10.28 ms (97.3 frames per second).
- React 18 Signal Intelligence Lab: Provides a dark obsidian chassis (#0B0F12) featuring: (1) HTML5 Canvas peak oscillogram with scrubbable audio playhead; (2) synchronized temporal self-attention heat-strip highlighting suspicious phoneme intervals; (3) dynamic 2D spectrogram toggleable between LFCC and Log-Mel; (4) calibrated confidence arc gauge with an explicit $[0.40, 0.60]$ uncertainty band; and (5) one-click printable forensic certificate via @media print.

## VII. Forensic Limitations and Ethical Disclosures

### A. Forensic Limitations and Failure Modes

1) Lossy Bitrate Smearing: Consumer compression below 16 kbps eliminates high-frequency phase incoher-

ence, shifting model outputs toward the uncertainty margin.

2) Short Duration Threshold: Utterances shorter than 1.0 s provide insufficient temporal frame context for BiGRU context propagation.
3) Unseen Diffusion Vocoders: Diffusion vocoders unrepresented in ASVspoof 2019 training data exhibit distinct acoustic signatures that require multi-domain calibration.

### B. Ethical Boundaries and Defensive Mission

Acoustic Forensics is designed strictly for defensive media authentication and digital forensics. Because countermeasure predictions are probabilistic estimates rather than absolute legal proof, outputs in high-stakes legal investigations must be verified by certified forensic examiners alongside secondary metadata.

## VIII. Conclusion and Future Research

We presented Acoustic Forensics, a comprehensive synthetic speech detection framework. By tracing the progression from classical statistical baselines (11.46%–23.94% EER) to an attention-pooled CRNN-LFCC countermeasure, we achieved 6.70% Eval EER on ASVspoof 2019 LA. We documented the critical empirical finding that score-level stacked generalization overfits development distributions (10.74% Eval EER), establishing standalone CRNN-LFCC as the superior production system. Extensive auditing exposed silence shortcut bias and codec degradation, providing actionable engineering constraints for real-world deployment. Future research will explore self-supervised representations (WavLM, HuBERT), acoustic room impulse response (RIR) augmentation, and multi-domain channel normalization.

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